St Mary's DSG, Kloof

Grade 10 | Physical Science Test
Mole calculations | 18 May 2022

TIME: 30 min | MARKS: 26

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NAME: MEMO: _____

SCIENCE TEACHER: _____

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 3 pages.
   A data sheet is provided.
   Please check that your question paper is complete.

2. Answer on your answer sheet in the spaces provided.

Question 1

Consider a 201,6 dm^3 sample of carbon dioxide (CO_2) gas at STP. Calculate:

1.1 The number of moles of CO_2. (2)

$n = \frac{V}{V_m} = \frac{201,6}{22,4} = 9\ mol$

1.2 The number of molecules present in this sample of CO_2. (2)

$N_0 = n \times N_A$
$= 9 \times 6,02 \times 10^{23}$
$= 5,42 \times 10^{24}$

1.3 The molar mass of CO_2. (2)

$M_r(CO_2) = 12 + 2(16)$
$= 44\ g.mol^{-1}$

1.4 The mass of CO_2. (2)

$n = \frac{m}{M} \quad m = n \times M_r = 9 \times 44 = 396\ g$

[8]

Question 2

2.1 The balanced equation for the decomposition of ammonium nitrate is given below:

$$NH_4NO_{3(s)} \longrightarrow N_2O_{(g)} + 2H_2O_{(g)}$$

2.1.1 In one such decomposition 3,01 x 10^{24} molecules of water are released. Calculate the number of moles of water released. (2)

$n = \frac{N_0}{N_A} = \frac{3,01 \times 10^{24}}{6,02 \times 10^{23}} = 5\ mol$

2.1.2 Calculate the number of moles in 240 g of NH_4NO_3. (3)

$M_r(NH_4NO_3) = 14 + 4(1) + 14 + 3(16)$
$= 80\ g.mol^{-1}$

$n = \frac{m}{M_r} = \frac{240}{80} = 3\ mol$

2.1.3 Calculate the volume of 4 moles of N_2O at STP. (2)

$V = n \times V_m$
$= 4 \times 22,4$
$= 89,6\ dm^3$

2.2 Ammonia gas is prepared in industry from the reaction of hydrogen gas and nitrogen gas, in the presence of a catalyst, as shown in the equation;

$$N_{2(g)} + 3H_{2(g)} \leftrightarrows 2NH_{3(g)}$$

Calc. mdes

2.2.1 ~~What volume~~ of hydrogen is needed to completely react with 1,5 moles of nitrogen? (3)

$H_2 : N_2$
$3 : 1$
$x : 1,5$
$x = 4,5\ mol$

2.2.2 What mass of ammonia is formed when 16,8 g of nitrogen reacts with excess hydrogen? ~~(4)~~ (5)

$n_{N_2} = \frac{m}{M_r} = \frac{16,8}{28}$
$= 0,6\ mol$
$N_2 : NH_3$
$1 : 2$
$0,6 : x$ m
$x = 1,2\ mol$

$M_r(N_2) = 28\ g.mol^{-1}$
$M_r(NH_3) = 14 + 3(1)$
$= 17\ g.mol^{-1}$
$m_{NH_3} = n \times M_r$
$= 1,2 \times 17$
$= 20,4\ g$

2.2.3 What volume of hydrogen is needed to form 7,84 dm^3 of ammonia at STP? (4)

$n_{NH_3} = \frac{V}{V_m} = \frac{7,84}{22,4} = 0,35\ mol$

$NH_3 : H_2$
$2 : 3$
$0,35 : x$ m
$x = 0,525$

~~V~~ $V = n \times V_m$
$= 0,525 \times 22,4$
$= 11,76\ dm^3$

[18]